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Echo State Network Optimization: A Systematic Literature Review

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Echo State Network Optimization: A Systematic Literature Review

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Abstract

In the recent years, numerous studies have demonstrated the importance and efficiency of Reservoir Computing (RC) approaches. The choice of parameters and architecture in reservoir computing, on the other hand, frequently leads to an optimization task. This paper attempts to present an overview of the related work on Echo State Network (ESN) optimization and to collect research papers through a systematic Literature Review (SLR). This review covers 74 items published since 2004 and 2021 that are concerned with the issue of our focus. The collected papers are selected, analysed and reported. The results indicate that there are two techniques of parameters optimization (bio-inspired and non-bio-inspired methods) have been extensively used for various reasons. But the bio-inspired ones are the greatest. The potential use of Particle Swarm Optimization (PSO) has also been noted. A significant portion of the research done in this field focuses on the study of reservoirs and how they behave in relation to their unique qualities. In order to test reservoirs with varied parameters, topologies, or training techniques, NARMA, the Mackey glass, and Lorenz time-series prediction dataset are the most commonly employed in the literature. This research debate diverse point of view about ESN's hyper-parameter optimization, metrics, datasets, evaluation measures, and bio-inspired and non-bio-inspired techniques, this paper identifies and explores a number of research gaps.

Keywords: Echo State Network; methods; Optimization; parameters; Reservoir Computing; SLR.

1. Introduction

In the last decades the use of Reservoir Computing is increasing [1] especially with the efficiency of the Echo State Network. The ESN [2] is new variant of Recurrent Neural Networks (RNN). The name of ESN springs from the input history echoes. These echoes represent the reservoir activation states (randomly generated), which include the input history. The ESN is distinguished from the other RNN by its hidden layer that is a large-scale sparsely connected reservoir and its training process that allows training the connection weights based on linear regression method [3]. ESN is known by simple structure and high prediction precision. Accordingly, this approach allows training RNN with competitive computational capability [3]. This training through ESN needs experience as the randomly created reservoir is relied on the defined global parameters. These parameters have to be defined correctly for gaining high results. To boost ESN performance, researchers have optimized a number of parameters [4] to increase efficiency.

In this paper, we intend to undertake a Systematic Literature Review that includes all of the most often cited and most recent strategies for improving ESN. The motivation of this review is to conduct a coherent study of ESN optimization, to give a clear statement of related works, to interpret, to discuss them and to provide baselines for future research.

The following are the significant distinctions between our SLR and other recent surveys:

1. The first systematic literature review in this topic, as far as we know,
2. In-depth list of covered related works on optimized echo state network,
3. The main scope of the SLR, where the most of the related papers is recently published,
4. A classification and analysis of the selected papers.

This paper describes first the theoretical background on Reservoir Computing and echo state network. Then, it presents the followed methodology to select the papers dealing with the main topic of this paper. The outcomes of the data extraction and interpretation are then presented. Following, it provides synthesis discussions. Finally, it concludes and presents research future suggestions.

2. Background

2.1. Echo State Network

ESN is a powerful variant of Recurrent Neural Networks (RNN) [5]. Like all RNN, there are three layers of ESN: the input, hidden, and output layer. The key idea of ESN is its very simple way to learn. It is a feedforward network that makes the early layer random and fixed. Training the last layer, which is a simple linear regression, is used for transforming input to predict output.

The rapidity of the ESN is due to the fact that only the connection weights between the hidden and output layers must be trained during the training process. The learning is then very fast. It is important also to set carefully random connection so that ESN do not die or explode. The ESN's structure is given in Fig. 1. The hidden layer in ESN is a big reservoir layer. From the input layer to the reserve pool layer, the connection weights are Win, where W is the connection weights of the reservoir neurons. And from hidden layer to output layer the connection weights is W_{out} (the name for bias isW_{out}^{bias}). A W_{back} connection from the output nodes to the reservoir (shown dotted in Fig.1) is optional. ESN can make multi-step predictions when W_{back} is employed. However, sometimes a single-step forecast is sufficient. The time step t is equivalent to u(t), and the number of neurons in the reservoir layer is equivalent to Nr. The Nr neuron's reservoir state is x(t). In the output layer, the output activity of L neurons is y(t). Each time step u is entered, the reservoir states are updated.

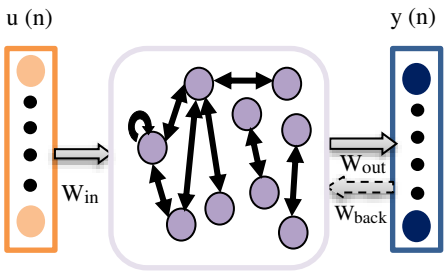


Fig. 1. Echo State Network structure (adapted from [10])

As shown in equations (1) and (2), the internal state of the ESN neurons is updating. The activation function for hidden layers is f. To handle the transmission of the input signal from the input layer to the reserve pool, f is typically a nonlinear function such as *tanh* or *sigmoid*. The x(n) is the reservoir's internal updating state, and y(n) is the output of the linear function f_{out}.

$$x(n+1) = f(Wx(n) + W_{in}\, u(n+1) + W_{back} y(n)) \quad (1)$$

$$y(n+1) = f_{out}\,(W_{out}[x(t+1), u(t+1), y(t)] + W_{out}^{bias}) \quad (2)$$

2.2. Parameters Optimization

The structure of ESN is shown in Fig. 1 below, with Nu neurons in the input layer, Nr neurons in the reservoir, and Ny neurons in the output unit. W is the state weights matrix (Nr*Nr) of the neuron in the reservoir, and Win is the weight matrix (Nr*Nu). The weight matrix (Nr*Ny) is called W_{out}.

The Spectral Radius (SR) [6], which defines the absolute value of the maximum Eigenvalues of the W matrix of the reservoir, is another discriminant parameter in Echo State Networks. This parameter gives an indication of an Echo State Network's short-term memory performance. Also we can note that a large value of spectral radius means slow descending of impulse response. Moreover SR may mirror the number of nonlinear interaction of input units through time so that a

big value of spectral radius means extended interactions. Researches have recommended a 0<SR<1, that can guarantee that ESN preserve the echo state properties.

Not only SR is an important parameter to optimize but also the scale factor IS which is multiplied by the input signal when it fed into the reserve pool. In general, this IS explaining the amount of the nonlinearity of an ESN.

The proportion of connectivity between neurons is represented by the Sparsity Degree (SD) [7] of the inner weight matrix W, which represents the link into the reservoir layer. If SD=100%, the ESN becomes a typical RNN. However, to lower the cost of updating the state of reservoir neurons, the ESN must use sparsely connected reservoir neurons (1). The most common choices of sparsity degree [8] used in the literature are between 2% and 10%.

To summarize, the reservoir size, spectral radius, input scaling, and sparsity matrix all have a significant impact on the ESN's performance. Furthermore, studies show that the parameters have no relationship [IE9]. Therefore, the ability to effectively and precisely tune the parameters that allow ESN to manner nonlinear systems to achieve optimal performance has become the modern-day research battlefield.

3. Search strategy

The search strategy aims at presenting the Systematic Literature review of ESN parameters optimization techniques. Many strategies have been proposed to classify and summarize collected papers from different digital databases.

The methodology followed by this survey is based on the guidelines of [9]. It aims at analyzing the methods of optimization of an ESN. Our conducted systematic method is divided into three main phases: (1) the aim and research questions, (2) the method of finding and selecting relevant studies, and (3) the knowledge extraction from these publications.

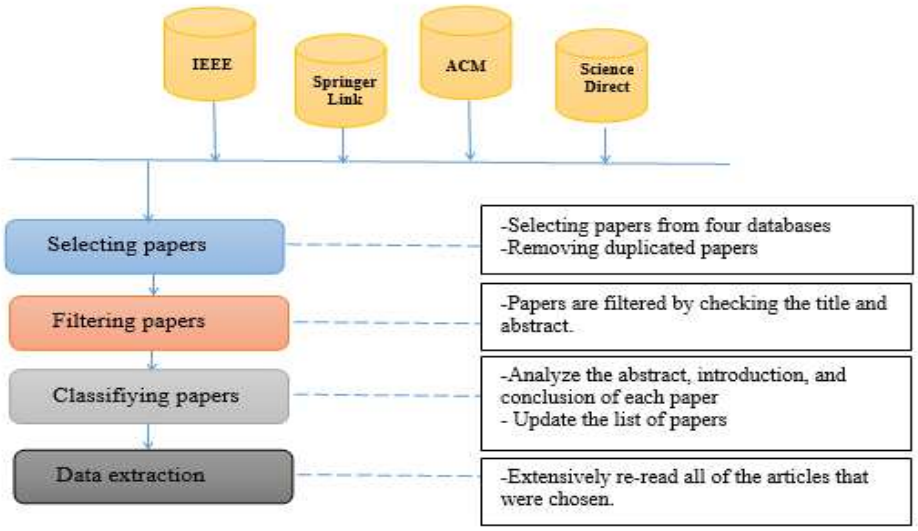


Fig. 2. Article Distribution by digital databases

A systematic literature review on how to optimize an echo state network was conducted. The research questions are defined as the initial stage of the selecting procedure. Following are the research questions we have come up with:

- RQ1: How far has this field of study progressed?
- RQ2: What are the most frequent methods in optimizing an Echo State Network?
- RQ3: what are the commonalities between the different proposed techniques of optimization?
- RQ4: What are the most frequent datasets used in the proposed algorithms?

3.1. Selection process

3.1.1. Search

The search phase consists on collecting the relevant papers focusing on optimized Echo State Network. We chose a set of proper search strings based on this selection, with which we expected to find the majority of the linked scientific articles. We have conducted a manual search throughout the following five digital databases: IEEEXplore Digital Library, SpringerLink Digital Library, ACM Digital Library, and ScienceDirect. In this survey, we have concentrated on indexed journals and conference proceedings. To retrieve papers from these databases, we have applied the following search command:

```
("Echo State Network" OR "ESN") AND ("optimization" OR "optimized")
```

Before proceeding to the next step, we have eliminating duplicates papers. The output of this step gives rise to 121 articles that were selected for being related to Echo State Network and its optimization. In our case, after the backward snowballing, 65 papers were collected and were included in the result (total of 950).

3.1.2. Criteria for inclusion and exclusion

We specified criteria for inclusion and exclusion in this step, as indicated in Table 1. Based on these criteria our electronic search was refined.

Table 1. Inclusion and Exclusion criteria

Selection criteria			
Inclusion criteria		**Exclusion criteria**	
I1	Only papers about ESN optimization should be included.	**E1**	Remove any documents that did not include an optimized ESN.
I2	Only studies that were published after 1st January 2004 are included.	**E2**	All papers published before 1st January 2004 are excluded.
I3	Only one copy of a duplicated research article should be included.	**E3**	Remove any copies of a duplicated research article that remain.
I4	Include just English-language research.	**E4**	All items published in a foreign language should be excluded.
I5	Only original research articles, full papers, and short papers of at least 5 pages should be included.	**E5**	Remove all articles of grey literature from the equation (i.e., abstracts, posters, technical reports, and work in progress papers).
I6	Only the most recent model version of the same authorship should be included.	**E6**	Remove any remaining things that are not needed (i.e., research papers of the same authoring and research subject).
I7	Include only items that can be downloaded (i.e., license grant).	**E7**	Remove any items that are not available for download (e.g., those without the subscription or copyright license).

First, we used a set of criteria to pick research papers that contained enough data to address our review issue: ESN optimization. With applying I1 and E1 we have remarked that many several research papers on Echo State Network do not introduce optimization algorithms. In the second step (I2, E2) of selection we focused on paper published between 2004 and 2021 because since 2004, all other nonlinear dynamical modelling methods have been outperformed by ESNs.
Other Inclusion and Exclusion criteria (I3, I4, I5, I6, I7, E3, E4, E5, E6 and E7) allow eliminating duplicated, non-English, redundant and non-downloadable research items.

In total, 950 research publications were found using the five electronic search engines. After applying inclusion and exclusion criteria, we kept 121 papers (Table 2 provides more information.).

Table 2. Selection of papers

	Results of the search command	Papers number after applying inclusion criteria	Papers number after applying exclusion criteria
IEEE Xplore Digital Library	353	70	**35**
ACM Digital Library	44	27	**15**
Springer	55	28	**15**
ScienceDirect	498	101	**56**
Total	950	226	121

The results of filtering process are presented in Table 3. First, we went over all of the titles and abstracts. Then, using three criteria, we categorized each paper: "potential," "doubt," and "out."

There were 121 research articles left for the second round of selection at that date. After reading the introduction and conclusion sections, only 92 papers are chosen. Finally, we read all of the full manuscripts before selecting 74 papers.

Table 3. Filtering of papers

Filtering Phase **Electronic search engines**	**1st filtering phase (Title and Abstract)**	**2nd filtering phase (Introduction and Conclusion)**	**3rd filtering phase (Full-text)**
IEEEXplore Digital Library	35	31	29
ACM Digital Library	15	11	4
SpringerLink	15	15	15
ScienceDirect	56	40	26
Total	121	97	74

4. Data Extraction

In this section, we will go over the steps for extracting data from the selected objects and analysing it to answer the SLR's research questions.

We used a data extraction form to extract the needed data from the selected papers, which included the data items we thought would be needed to address the review's research questions.

Each item must have at least one term in each group whose aim is to be graded on the list of classified papers.

Consequently, 74 articles have been categorized, as summarized in Table 4.

We classified our analysed papers into two groups of key goals:

- The first group: deals with bio-inspired optimization methods.
- The second group: focus on non-bio-inspired optimization methods.

The first group of studies comprised those that used biology as a source of inspiration and were known to be effective at solving such difficulties. This collection of papers focuses on aspects of the Echo State Network where biology is applied (ESN). The papers that underlined non-bio-inspired methods fall into the second category.

For each group we have extracted the following data:

(1) Datasets used: What is the dataset used?
(2) Methods: what are the methods of optimization applied?
(3) Parameters and Topology: At what level is optimization of ESN is applied?
(4) Objective function: what are the metrics of evaluation in the studied researches?
(5) Comparison: what is the added value compared to such method?
(6) Results

We provide the findings of the data analysis and classification in the next section based on gathered data.

Table 4. Distribution of selected papers by category.

	Bio-inspired methods	Non bio-inspired methods	Total
IEEEXplore Digital Library	[IE1] [IE2] [IE3] [IE4] [IE5] [IE6] [IE8] [IE9] [IE10] [IE11] [IE12] [IE13] [IE15] [IE16] [IE17] [IE18] [IE21][IE22] [IE23] [IE24] [IE25] [IE26] [IE28]	[IE7] [IE9] [IE14] [IE19] [IE20] [IE27] [IE29]	
	Total: 23	**Total:** 7	**30**
ACM Digital Library	[AC2][AC3]	[AC1][AC4]	
	Total: 2	**Total:** 2	**4**
ScienceDirect	[SD1][SD3] [SD6] [SD7][SD8] [SD9] [SD10] [SD13] [SD14] [SD15] [SD16] [SD18] [SD19] [SD21] [SD22] [SD23] [SD24] [SD25] [SD26]	[SD2] [SD4] [SD5] [SD11] [SD12] [SD17] [SD20]	
	Total: 19	**Total:** 7	**26**
SpringerLink	[SP3] [SP4] [SP5] [SP6] [SP7] [SP8] [SP9] [SP10] [SP12] [SP13] [SP14] [SP15]	[SP1] [SP2] [SP11]	
	Total: 12	**Total:** 3	**15**
Total	**56**	**19**	**75**

By analysing the extracted data, we can answer the previously defined research questions of our study (cf. section 3.1).The studies we chose have been grouped into two groups (cf. section 3.3): (1) bio-inspired optimization methods, and (2) on non-bio-inspired optimization methods. Table 4 shows the distribution of papers in each category as well as the number of papers in each category. Bio-inspired approaches have sparked the most interest, as can be seen.

4.1. RQ1: Status of the field

In the previous section, we have extracted data from the 74 selected papers. These papers are published in two different sources, including conferences and journals. Fig.2 presents the papers' types in each category and as well as the papers' number found in each type. The majority of the studies were published in journals, as evidenced by Fig.2.

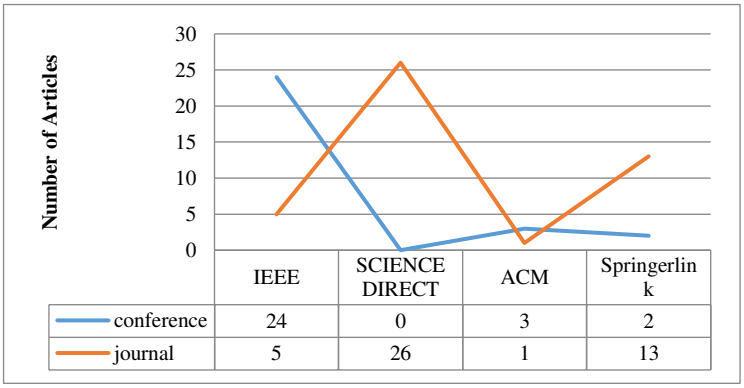


Fig. 3. Article Distribution by digital databases

In this SLR, papers are graded by the year of apparition between 2004 and 2021. As we can see, Echo State Networks are a type of recurrent neural network that was introduced around the beginning of the 2000s [11]. Since its apparition, the ESN overcomes the limitations during the training of the RNNs while no significant drawbacks are introduced. But the ESN model presents some clear-cut disadvantages when the parameters are not well initialized. Considering that 2007, the posted papers have end up common as compared to preceding years. The years 2011 may be meant as critical milestones because of the growth of posted articles. Additionally, the quantity of works posted in 2018, 2019, 2020 and early 2021 suggest the developing hobby of this area till now (cf. Fig. 3). We are aware that ESN optimization algorithms were attracting greater interest in the future with greater posted works in specialised meetings and unique troubles of journals.

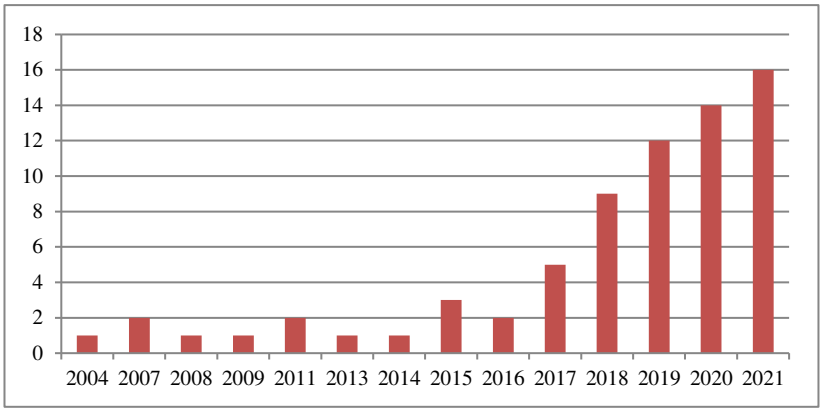


Fig. 4. Distribution of number of papers by year of publishing

4.2. RQ2: the most frequent methods in optimizing an echo state network

To answer this research question, from the reviewed studies (defined in section 3.3), table 5 presents the captured data about RQ2 for each category.

Table 5. Distribution of selected papers by used techniques.

	Methods	References of the papers	Total
Bio-inspired methods	Particle Swarm Optimization (PSO)	[IE8] [IE10] [IE11] [IE12] [IE16] [IE18] [IE19] [IE23] [IE24] [SD1] [SD7] [SD10] [SD19] [SD21] [SP5][SP15]	16
	Genetic algorithm (GA)	[IE1] [IE9] [IE17] [IE21] [SD6] [SD13] [SD14] [SD25] [SP5] [AC1]	10
	Differential Evolution (ED)	[IE21] [SD9] [SD18] [SP5] [SP9]	5
	Evolutionary Strategy (ES)	[IE2] [IE4] [SD16] [SP10] [SP12]	5
	Backtracking Search Optimization	[IE5] [IE25] [SD3]	3
	Cuckoo Search Optimization	[IE6] [SP4]	2
	Fruit Fly Optimization Algorithm	[SD23] [SP3]	2
	Grey Wolf Optimizer	[SD8] [SP7]	2
	Intrinsic Plasticity	[IE13] [SD24] [SD26]	2
	Quantum-Behaved Fruit Fly Optimization	[IE15]	1
	Adaptive Echo Etate Network	[SD22]	1
	Anti-Oja's Learning	[SP6]	1
	Binary Grey Wolf Optimization	[SD15]	1
	Microbial Genetic Algorithm	[IE26]	1
	Bacterial Foraging Algorithm	[IE28]	1
	Bat Optimization	[IE22]	1
	Grasshopper Optimization Algorithm	[IE3] [SP7]	2
	Whale Optimization Algorithm	[SP11][SP13][SP14]	3
Non-bio-inspired method	Bayesian Optimization	[IE7] [SD17] [AC1]	3
	Gradient Descent	[SD12] [SD20]	2
	Cross-validation method	[IE9] [SP11]	2
	State transition algorithm	[SD11]	1
	Fractional-order optimization	[SP2]	1
	Gravitational Search optimization	[SP1]	1
	Kronecker product	[IE21]	1
	Mini-batch optimization	[AC2]	1

Multi Reservoir	[IE20]	1
Newton method	[SD2]	1
Scenario Approach	[SD5]	1
least absolute shrinkage and selection operator	[IE29]	1
Broyden–Fletcher–Goldfarb–Shanno quasi-Newton algorithm	[IE27]	1
Seasonal Adjustment Method	[SP8]	1
Stochastic Gradient Descent	[SD4]	1
Recurrent architecture	[SP10]	1
Contribution reservoir neurons	[IE14]	1

The Echo State Network was created to solve the difficulty of Recurrent Neural Networks' gradient-descent training [12]. The selection of ESN parameters, on the other hand, frequently requires optimization. Several methods are used to optimize Echo State Network in order to increase its performance.

In this context, this analysis was done, using extraction techniques. Table 5 was used to visualize the scores of each method of optimization. From table 5 we can conclude that Metaheuristics (MA) techniques [13] have the potential to be used to improve ESN's parameters [14]. Several studies demonstrate that the use of MAs methods (using mathematical features) leads to a higher accuracy and convergence rate than other analogues. Besides, MA's or Evolutionary algorithms are known to have not too much tuning parameters; due to this fact their implementations are quite simple.

By analysing Table 5 and Fig.5 we can conclude that Particle Swarm Optimization is the most used MA's techniques. Research has shown that this relatively young optimization method has a very short computation time and it is very easy to program all these factors make it powerful. PSO employs a swarm of agents (particles) that moves through the research area looking for the optimal solution. Each particle in the research space changes its "flight" based on its own and other particles' previous flight experiences. Each particle leaves a trail: its best answer (personal best), any particle's best value, global best, and gbest. The changing velocity in each iteration affects the new position. [15] Provides an overview of the PSO approaches.

Not only PSO is potentially used but also, other evolutionary computation techniques [16] were found to be competitive with other techniques for optimization ESN's parameters. Genetic algorithms are among these algorithms [17]. They are optimization algorithms based on genetic and natural evolution techniques: crosses, mutations, selection, etc. Genetic algorithms already have a relatively old history since the early work of J.H. Holland in 1992 [17]. A genetic algorithm searches for the optimum(s) of a function defined on a data space.

To make use of it, we need the following five things: (1) The coding principle for the population element. (2) A method for populating the starting population. This process must be able to generate a population of non-homogeneous individuals that can be used as a foundation for future generations. (3) A function that has to be improved. This yields a value known as the individual's fitness or evaluation function. (4) Operators that allow for population diversification across generations and space exploration. The crossbreed operator recomposes the genes of existing individuals in the population, while the mutation operator ensures state space exploration and (5) Parameters for dimensioning: population size, total number of generations or criterion stop, application probabilities of crossover and mutation operators.

Evolution Strategy, Differential Evolution, backtracking Search, and other bio-inspired-techniques are used also to optimize parameters of the ESN.

Another intriguing result is that some studies have shown that intrinsic plasticity can be used to enhance synaptic connections. The most prevalent brain plasticity processes are synaptic plasticity [18] and intrinsic plasticity [19], which characterizes a neuron's ability to change the type of response it gives to the same input. The intrinsic properties of the neuron or the properties of synapses can be affected by changes. In the first case, it is intrinsic plasticity and in the second case, it is synaptic plasticity. Neuronal plasticity constitutes the unitary mechanism of memorization and learning.

In the other side of techniques (non-bio-inspired techniques), we can note that the Bayesian optimization (BO) have the potential to be used. BO [20] is a one-sided surrogate-based optimization technique that uses a gradient-free global optimization technique to optimize arbitrary

functions [21]. When gradients are unreachable, it was introduced to reduce loss functions of an artificial neuron network whose (hyper)parameters are difficult to modify, or when objective function is a non-convex one (many local optima). Additionally, Table 5 indicates remarkable presence of gradient descent algorithms [22] with its two variants (gradient descent and stochastic gradient descent).

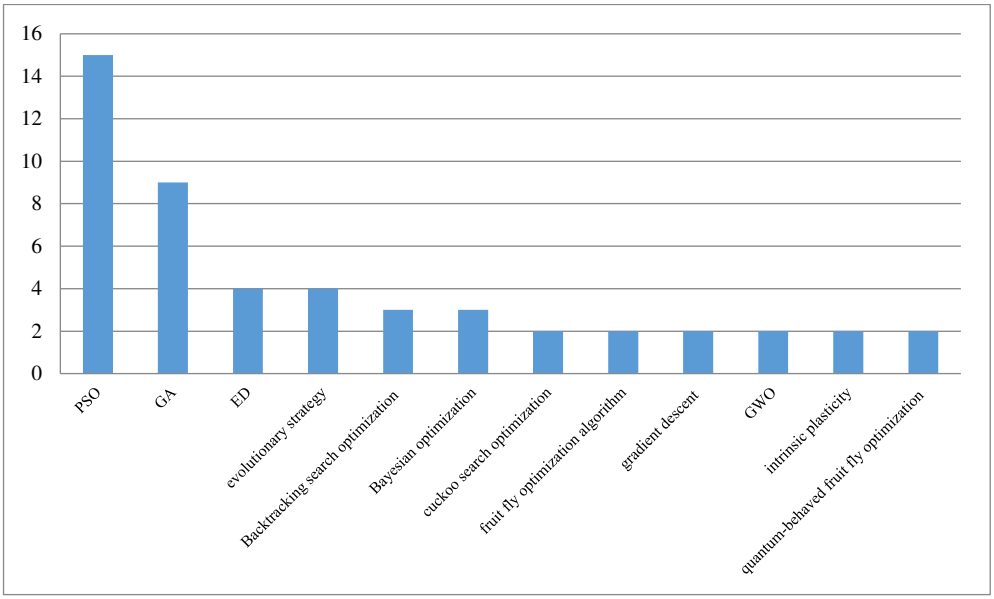


Fig. 5. Distribution of optimization methods by studied papers

4.3. RQ3: The commonalities between the different proposed techniques of optimization

To answer this research question, we have extracted data (3) and (4) from the reviewed studies (defined in section 3.3). Table 6 presents the captured data about RQ3 for each category.

Table 6 . Commonalities.

	1.1.1. Global parameters (reservoir, the spectral radius, and the scaling of weights)	[1] **Topology optimization (connectivity, weights)**
Reference papers	[IE1][IE2][IE3][IE4][IE6][IE7][IE9][IE1] [IE15][IE18][IE23][IE25] [IE26][IE27][IE28]	[IE1][IE2][IE3][IE5][IE6][IE7][IE8][IE10] [E11] [IE12] [IE13] [IE14] [IE16] [IE20][IE22] [IE24] [IE29]
	[SD1][SD2][SD6][SD7][SD8][SD9] [SD11][SD12] [SD13][SD14][SD17][SD18][SD20] [SD21][SD24] [SD25]	[SD3][SD5][SD7][SD8][SD9][SD10] [SD11] [SD15][SD16][SD18][SD22][SD24] [SD26]
	[SP2][SP3][SP4][SP9][SP12] [SP13] [SP14] [SP15]	[SP1][SP2][SP4][SP6][SP7][SP10]
	[AC1][AC2][AC3]	[AC2][AC3]
Process	[2] The first class of articles optimizes the reservoir size, spectral radius, input/output scaling, input/output shift, type of readout function, regularization parameter of ridge regression, reservoir activation function, leaking rate, and noise scaling. [3] In this scenario, the parameter representation can be a 1-dimensional vector of sufficient length to fit the hyper parameters to be optimized. [P1, P2, ..., Pn], for example, where P is the parameter to be optimized and denotes the number of parameters to be optimized.	The second group of studies focuses on improving the network's topology, such as weight connections and reservoir connectivity. In this Case, weights can be a two-dimentional binary matrix.where lines and colums of matrix represent neurons.for example in a matrix C ,C_{ij} denotes connection between $neuron_i$ and neuron $_j$ .if the value of C_{ij} is 1, this indicates the presence of a link between neuron I and neuron j, whereas (Cij=0) indicates the absence of such a connection.
Objective function	ESN offer a significant advantage over other forms of RNNs in terms of their ability to learn in a single step. Only Wout weights are used in the training. As a result, ESN training is typically quick. Those expressing ESN prediction accuracy, such as Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Normalized RMSE (NRMSE), etc, are the most commonly used objective functions.	

4.4. RQ4: the most frequent datasets used in the proposed algorithms

In this section, we focused on the most used datasets in the studied papers.

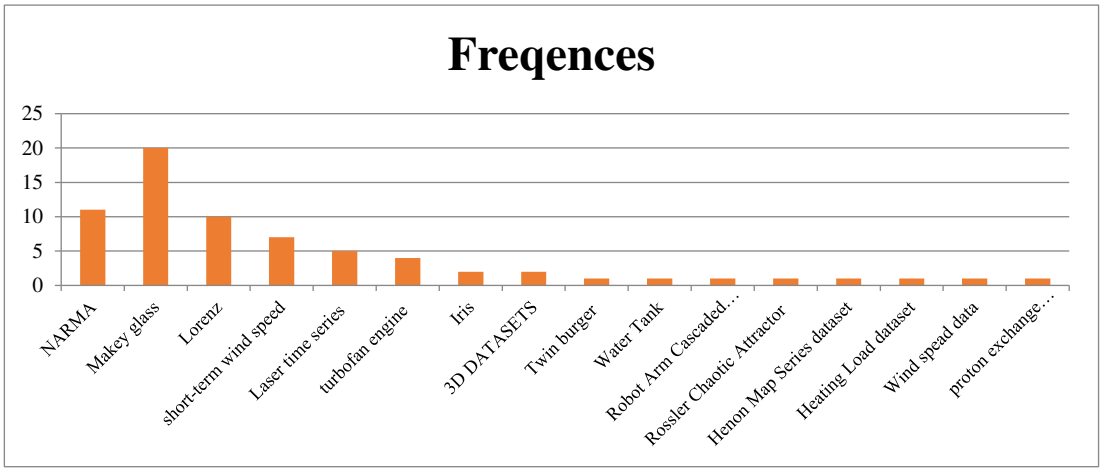


Fig. 6. Distribution of some datasets by studied papers

(1) NARMA and (2) Mackey Glass: the two most frequent datasets in our studied papers are NARMA and Mackey glass which are two real data benchmarking sequences. They are characterized by their nonlinear autoregressive moving average [23]. One of the most often used called NARMA which is characterized by its unpredictable sequence complexity and its high stages of misunderstanding and non-correlation of inputs. Its dynamic expression is given by the following equation:

$$y(t+1)=c_1y(t)+c_2y(t)\ \sum_{i=0}^{k-1}\ y(t-i)+c_3x(t-(k-1))x(t)+c4 \qquad (3)$$

The output and input of the system at time t are denoted by y(t) and x(t), respectively. Initial values are given to the constant parameters ci. The complexity of NARMA is determined by parameter k. In most cases, k is set to 10 or 30, which are the two most popular values in the literature.

(3) The Lorenz time-series prediction: a dataset which is used as a benchmark for ESN [8]. Equation (4) describes the Lorenz dynamic system:

$$\frac{dx}{dt} = a(-x + y);\ \frac{dy}{dt} = bx - y - xz;\ \frac{dz}{dt} = xy - cz \qquad (4)$$

The Lorenz system is a nonlinear equation system because of the terms xy and xz. This system cannot be integrated in the general case. The determination of this system must be done using approximation methods. When the real parameters σ, r and b take the following values:
a = 10, b = 28 and c = 8/3, with the initial conditions x (0) = y (0) = z (0) = 0.01, the system (2.9) is chaotic.
We can note also that many others benchmarks are used in this field. Some of them are private data such NASA's turbofan engine. As well as others that are publicly accessible, such as Iris.

5. Discussion

This paper presents a survey of recent researches that focuses on techniques of optimization of Echo State Network's parameters.

5.1. Findings

The most findings from this study are the following. In fact, ESN optimization algorithms were attracting greater interest in the course of the final years with huge number of posted items. Possibly, because ESN still a young field of research, moreover reservoir computing proves its efficiency in time series tasks.

Furthermore, bio-inspired approaches have been employed extensively to tune ESN parameters. The optimization parameters that have a considerable impact on the ESN's performance are size of the reservoir, connectivity rate, and spectral radius. The size of the reservoir is important because

it determines the capacities of the reservoir whether it is the computational or memory capacity. As with neural networks, too many units can lead to over-learning and waste of computing time. As a result, we can see that a reservoir's memory capacity cannot exceed its number of units; this must be considered in order to integrate enough units to match the task's requirements.

The connectivity rate is also an important parameter, which specifies the number of connections between the units of the reservoir. Strong connectivity indicates that the units are all connected to each other, while weak connectivity [12] indicates that each unit is only connected to a small number of other neurons. A low connectivity reservoir shows slightly better performance but does not affect much the capacities of a reservoir. However, a weakly connected reservoir has a matrix W made up of large part of zeros.

Thus, with a representation of the sparse matrices, a weakly connected reservoir allows to calculate more quickly the updates of its states. It is therefore advisable for this parameter to use a weakly connected reservoir (10%), regardless of its size, in order to save calculation time. the spectral radius is another decisive parameter which indicates the greatest eigenvalue in value absolute. A spectral radius less than unity (spectral radius < 1) is regarded a satisfying the following condition for the attribute of reasoning in a vast number of publications. Based on the studies reviewed, we may deduce that spectral radius optimization is computationally expensive, and that researchers have just recently discovered a way to optimize the reservoir using metaheuristics.

The PSO is the mostly used technique to optimize the ESN. This may be due to its little number of parameters in addition to the simplicity in implementation. The PSO is indeed multifaceted sufficient as the choice optimization algorithm for reservoir computing methods and especially echo state network.

Another remarkable discovery is the fact that vast majority of reviewed works uses at least one of four benchmark problems which are NARMA, Mackey Glass, Lorenz time-series prediction and short-term wind speed. In reality, these benchmark datasets have been routinely utilized to test time series prediction difficulties. The chaotic properties of their system, as well as its extended memory, make forecasting challenging [23]. Furthermore, these datasets are applied to quantify the ESN's learning abilities.

5.2. Researches gaps and future works

Observing the different performance of the optimization algorithm, the fundamental disadvantage of these reservoir optimization approaches, that we may conclude, is that they either use too many factors, such as optimizing the reservoir weight, or they only improve a few effective characteristics. In both circumstances, there are issues. When there are too many optimization parameters in the first scenario, the optimization process is costly to run, which is impractical in some cases. The performance of the reserve pool cannot be considerably enhanced in the second instance, where only a few effective parameters are modified.

Particular research in neuroscience and in Reservoir Computing focuses on intrinsic plasticity which represents the capacity of a network to adapt its units and parameters. Several types of plasticity exist; such as intrinsic plasticity which regulates the excitability of neurons and induces stable homeostatic effects in the dynamics of an ESN. Another type of plasticity, synaptic plasticity, modifies synaptic weights to regulate impulses received by a neuron. However, its application has remained largely unsuccessful. Finally, synaptic standardization appropriately modifies the levels of a neuron's incoming connections so that the aggregate of all neurons' activations remains consistent.

Several strategies are utilized to reduce unnecessary reservoir connection weights during training to improve ESN performance. The echo state property cannot be preserved by cutting operations on the reserve pool. Meanwhile, many academics are focusing on cutting redundant connections from the reservoir to the output layer to optimize the ESN structure. The most often used strategies for optimizing network structure are regularization methods. These techniques can improve the Echo State Network's generalization ability while simultaneously addressing the issue of overlearning. Nonetheless, based on the dynamical organization of reservoirs and the number of training patterns, determining the appropriate value for the regularization parameter is difficult.

Additionally, the evolutionary computation algorithms were employed to optimize the ESN structure. Particle swarm optimization and genetic algorithms, as described in section 3.2, are utilized to optimize the ESN's weight connection topologies. The evolutionary computation, on the

other hand, techniques are heuristics algorithms that prove a high computational complexity. As a result of this strategy, another question arises: what is the most efficient method for cutting duplicate connection weights and improving ESN performance?

In future research, we may look into more appropriate intelligent algorithms for optimizing the W_{out} of Echo State Network, because training the output weights of ESN using the traditional linear regression method causes the trained network to over fitting. The use MA‘s to optimize ESN architectures may solve the problem. But we remember that these techniques are heuristics whose objective is not the same than that of the exact methods. These methods, in particular, do not ensure the discovery of an optimal solution or the proof of the optimality of the solution identified - unless a lower bound of the cost function is known and the solution obtained meets this criterion.

6. Conclusion

Our SLR concerns ESN considered as a Recurrent Neural Network variant with a faster training speed because only the W_{out} is trained. To improve the ESN's performance, many research works proposed optimizing its parameters. Several algorithms have been introduced, but the best are the bio-inspired ones, considered the first in reservoir computing. Furthermore, we have highlighted the potential use of PSO which is particularly effective for nonlinear, continuous variable, integer or mixed optimization problems. The study of reservoirs and their performance according to their specific characteristics constitutes a broad part of the work done in this area. For this, test benches are commonly used in the literature in order to test reservoirs with different parameters, different topologies or which are trained with different methods, in order to estimate the impact. Three of the most frequently used tests in the literature are NARMA, Mackey glass and Lorenz time-series prediction. The most cornered research gaps are the impact of number of optimized parameters, whether plasticity of ESN effect its performance, the optimization of the structure of ESN, as well as the readout weights.

Appendix : Selected Papers

The list of 74 papers examined in this SLR, arranged by digital database, is as follows:

IEEE

[IE1] Paiva P, Vellasco M, Amaral J. (2018) Quantum-Inspired Optimization of Echo State Networks Applied to System Identification. *2018 IEEE Congress on Evolutionary Computation (CEC),* pp. 1--8.

[IE2] Ishii K, van der Zant T, Becanovic V,Ploger P (2004) Optimization of parameters of echo state network and its application to underwater robot. *SICE 2004 Annual Conference,* vol. 3, pp. 2800--2805.

[IE3] Bala A, Ismail I, Ibrahim R, Sait SM, Oliva D (2020) An improved grasshopper optimization algorithm based echo state network for predicting faults in airplane engines. *IEEE Access,* vol. 8, pp. 159773—159789.

[IE4] Liu K, Zhang J (2018) Optimization of Echo State Networks by Covariance Matrix Adaption Evolutionary Strategy. *2018 24th International Conference on Automation and Computing (ICAC),* n° %14, pp. 1--6.

[IE5] Wu S, Wang Z, Ling D (2019) Echo state network prediction based on backtracking search optimization algorithm. *2019 IEEE 3rd Information Technology, Networking, Electronic and Automation Control Conference (ITNEC),* pp. 661--664.

[IE6] Bala A, Ismail I, Ibrahim R (2018) Cuckoo search based optimization of echo state network for time series prediction. *2018 International Conference on Intelligent and Advanced System (ICIAS),* pp. 1--6.

[IE7] Maat JR, Gianniotis N, Protopapas P (2018) Efficient optimization of echo state networks for time series datasets. *2018 International Joint Conference on Neural Networks (IJCNN),* pp. 1--7.

[IE8] Basterrech S, Alba E, Snášel V (2014) An experimental analysis of the echo state network initialization using the particle swarm optimization. *2014 Sixth World Congress on Nature and Biologically Inspired Computing (NaBIC 2014),* pp. 214--219.

[IE9] Cai M, Fan X, Wang C, Gao L, Zhang X (2018) Research for parameters optimization of echo state network. *2018 11th International Symposium on Computational Intelligence and Design (ISCID),* vol. 2, pp. 177--180.

[IE10] Chouikhi N, Ammar B, Rokbani N, Alimi AM, Abraham A (2015) A hybrid approach based on particle swarm optimization for echo state network initialization. *2015 IEEE International Conference on Systems, Man, and Cybernetics,* pp. 2896--2901.

[IE11] Chouikhi N, Fdhila R, Ammar B, Rokbani N, Alimi AM (2016) Single-and multi-objective particle swarm optimization of reservoir structure in echo state network,» *2016 International Joint Conference on Neural Networks (IJCNN),* pp. 440--447, 2016.

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[IE13] Fan HT, Wang W, Jin Z (2017), Performance optimization of echo state networks through principal neuron reinforcement. *2017 International Joint Conference on Neural Networks (IJCNN),* pp. 1717--1723.

[IE14] Li D, Liu F, Qiao J, Li R (2018) Structure optimization for echo state network based on contribution. *Tsinghua Science and Technology,* vol. 24, n° %11, pp. 97--105.

[IE15] Han Y, Jing Y, Li K (2017) Multi-step prediction for the network traffic based on echo state network optimized by quantum-behaved fruit fly optimization algorithm,» *2017 29th Chinese Control And Decision Conference (CCDC),* pp. 2270—2274.

[IE16] Long J, Zhang S, Li C (2019) Evolving deep echo state networks for intelligent fault diagnosis,» *IEEE Transactions on Industrial Informatics,* vol. 16, n° %17, pp. 4928--4937, 2019.

[IE17] Lu Y, Liao Y, Xu L, Liu Y, Liu Y (2021) Laplacian Deep Echo State Network Optimized by Genetic Algorithm. *2021 IEEE International Conference on Information Communication and Software Engineering (ICICSE),* pp. 107--111.

[IE18] Luo Q, Zhou J, Sun Y, Simpson O (2020) Jointly optimized echo state network for short-term channel state information prediction of fading channel. *2020 International Wireless Communications and Mobile Computing (IWCMC),* pp. 1480--1484.

[IE19] Lv H, Song Y, Jia P (2011) Parameter optimization using PSO for ESN-based robotic belt grinding modeling. *2011 3rd International Workshop on Intelligent Systems and Applications,* pp. 1--4.

[IE20] Mezzi R, Morando S, Steiner NY, Péra MC, Hissel D, Larger L (2018) Multi-reservoir echo state network for proton exchange membrane fuel cell remaining useful life prediction. *IECON 2018-44th Annual Conference of the IEEE Industrial Electronics Society,* pp. 1872--1877.

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[IE23] Salah SB, Fliss I, Tagina M (2017) Echo state network and particle swarm optimization for prognostics of a complex system. *2017 IEEE/ACS 14th International Conference on Computer Systems and Applications (AICCSA),* pp. 1027--1034.

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[IE27] Yuan J, Chen W, Luo A , Huo C, Ma S (2021) Short Term Load Forecasting Based on BFA-DRESN Algorithm. *2021 IEEE 2nd International Conference on Big Data, Artificial Intelligence and Internet of Things Engineering (ICBAIE)* , pp. 114-118.

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[SD1] Chen Y, He Z, Shang Z, Li C, Li L, Xu M (2019) A novel combined model based on echo state network for multi-step ahead wind speed forecasting: A case study of NREL *Energy Conversion and Management,* vol. 179, pp. 13-29.

[SD2] Lun SX, Hu HF, Yao XS (2019) The modified sufficient conditions for echo state property and parameter optimization of leaky integrator echo state network. *Applied Soft Computing,* vol. 77, pp. 750-760.

[SD3] Wang Z, Zeng YR, Wang S, Wang L (2019) Optimizing echo state network with backtracking search optimization algorithm for time series forecasting. *Engineering Applications of Artificial Intelligence,* vol. 81, pp. 117-132.

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